

Week 9.

Confidence intervals (CIs)

1 One-sample confidence intervals for normal sample

1.1 CI for μ when σ is known

Example 1.1. Let the sample $(X_1, \dots, X_n)$ come from $N(\mu, \sigma^2)$ and let $0 < \alpha < 1$. Then $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$, so a level $1 - \alpha$ exact confidence interval for μ when σ is known:

$$\left(\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right). \quad (1)$$

1.2 CI for μ when σ is not known

Let the sample $(X_1, \dots, X_n)$ come from $N(\mu, \sigma^2)$ and let $0 < \alpha < 1$. If σ is not known, we plug in S for σ in the above. Then $\frac{\bar{X}_n - \mu}{S/\sqrt{n}}$ is not normally distributed therefore $z_{\alpha/2}$ is not accurate in (1) (but it is approximately accurate when the size of the sample is large enough). The random variable $\frac{\bar{X}_n - \mu}{S/\sqrt{n}}$ is **t** - distributed with $n - 1$ degrees of freedom (see the Appendix).

Example 1.2. Let the sample $(X_1, \dots, X_n)$ come from $N(\mu, \sigma^2)$ and let $0 < \alpha < 1$. Then a $1 - \alpha$ level CI for μ is

$$\left(\bar{X}_n - t_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{X}_n + t_{\alpha/2} \frac{S}{\sqrt{n}} \right). \quad (2)$$

where $t_{\alpha/2} = t_{\alpha/2}(n - 1)$ is the $\alpha/2$ upper quantile of the **t** - distribution with $n - 1$ degree of freedom.

1.3 CI for σ^2

Example 1.3. Let the sample $(X_1, \dots, X_n)$ come from $N(\mu, \sigma^2)$ and let $0 < \alpha < 1$. Then by Theorem A.6, the random variable $\frac{(n-1)S^2}{\sigma^2}$ is chi-squared distributed with $n - 1$ degrees of freedom, thus

$$\mathbb{P} \left(\chi_{1-\alpha/2}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\alpha/2}^2 \right) = 1 - \alpha,$$

where $\chi_{\alpha/2}^2 = \chi_{\alpha/2}^2(n - 1)$, $\chi_{1-\alpha/2}^2 = \chi_{1-\alpha/2}^2(n - 1)$ are the $\alpha/2, 1 - \alpha/2$ upper quantiles of the χ^2 - distribution with $n - 1$ degree of freedom, respectively. Therefore, a $1 - \alpha$ level CI for σ^2 is

$$\left(\frac{(n-1)S^2}{\chi_{\alpha/2}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2} \right). \quad (3)$$

For example, if $\alpha = 0.05$, $s = 1.6$, $n = 20$ then a level $1 - \alpha$ CI for σ^2 is (1.48056, 5.46117) and for σ is (1.21678, 2.33691).

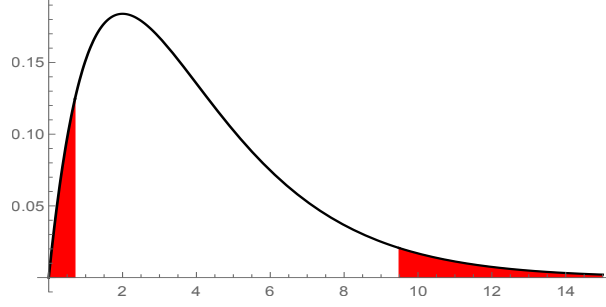


Figure 1: The $\chi^2_{1-\alpha/2}$ and $\chi^2_{\alpha/2}$ upper quantiles of a χ^2 distribution.



Figure 2: William Sealy Gosset

2 Two-sample confidence intervals

2.1 Large samples

Suppose that two large independent samples $(X_1, \dots, X_{n_1})$ and $(Y_1, \dots, Y_{n_2})$ are given and let $\mathbb{E}X_1 = \mu_1$, $\mathbb{E}Y_1 = \mu_2$, $\text{Var}X_1 = \sigma_1^2$, $\text{Var}Y_1 = \sigma_2^2$. Similarly to the one-sample case we have

$$\frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \xrightarrow{d} N(0, 1) \quad (4)$$

therefore when σ_1, σ_2 are known then a level $1 - \alpha$ *approximate* CI for $\mu_1 - \mu_2$ is

$$\left((\bar{X}_{n_1} - \bar{Y}_{n_2}) - z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right). \quad (5)$$

If S_1^2 and S_2^2 denote the sample variances of the $X_1, \dots, X_{n_1}$ and $Y_1, \dots, Y_{n_2}$ samples, respectively then it can be proven (by a more general version of the CLT) that

$$\frac{(\bar{X}_{n_1} - \bar{Y}_{n_2}) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \xrightarrow{d} N(0, 1),$$

therefore

$$\left((\bar{X}_{n_1} - \bar{Y}_{n_2}) - z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + z_{\alpha/2} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \right) \quad (6)$$

is also a level $1 - \alpha$ *approximate* CI for $\mu_1 - \mu_2$. Therefore (6) can be used when σ_1, σ_2 are not known and the sample sizes are large (textbooks recommend that the sizes be greater than 30).

2.2 Small samples when the parent distributions are normal

Example 2.1. Let the samples $(X_1, \dots, X_{n_1})$, $(Y_1, \dots, Y_{n_2})$ come from $N(\mu_1, \sigma_1^2)$, $N(\mu_2, \sigma_2^2)$ distributions (the so-called parent or underlying distributions), respectively, and let $0 < \alpha < 1$. Then if σ_1 and σ_2 are known then a level $1 - \alpha$ *exact* CI for $\mu_1 - \mu_2$ is given by (5), since in (4) the rv is in fact standard normally distributed.

If σ_1, σ_2 are not known, but it is known that $\sigma_1 = \sigma_2$ then a $1 - \alpha$ level *exact* CI for $\mu_1 - \mu_2$ is given by

$$\left((\bar{X}_{n_1} - \bar{Y}_{n_2}) - t_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X}_{n_1} - \bar{Y}_{n_2}) + t_{\alpha/2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right) \quad (7)$$

where $t_{\alpha/2} = t_{\alpha/2}(n_1 + n_2 - 2)$ is the $\alpha/2$ upper quantile of the t - distribution with $n_1 + n_2 - 2$ degree of freedom and $S_p = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$ is called the "pooled sample standard deviation" (and S_p^2 is called the "pooled sample variance") of the two samples.

Proof. Apply Theorem A.8.

3 The F - distribution

Definition 3.1. If $U \sim \chi^2(m)$ and $V \sim \chi^2(n)$ (See Definition A.4) independent then the rv $F = \frac{U/m}{V/n}$ is called a Fisher - distribution, or F - distribution with (m, n) degrees of freedom. We denote this by $F \sim F(m, n)$.

Proposition 3.2. Let $F \sim F(m, n)$, and for any $0 < \alpha < 1$ define $f_\alpha(m, n)$ as $\mathbb{P}(F > f_\alpha(m, n)) = \alpha$. Then $f_{1-\alpha}(n, m) = \frac{1}{f_\alpha(m, n)}$.

Proof. $\alpha = \mathbb{P}(F > f_\alpha(m, n)) = \mathbb{P}(\frac{1}{F} < \frac{1}{f_\alpha(m, n)}) = 1 - \mathbb{P}(\frac{1}{F} \geq \frac{1}{f_\alpha(m, n)})$ and we obtain the statement of the proposition by noting that $\frac{1}{F} \sim F(n, m)$.

Proposition 3.3. Let $(X_1, \dots, X_{n_1})$, $(Y_1, \dots, Y_{n_2})$ be independent samples with parent distributions $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively. Then

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1),$$

with S_1^2 and S_2^2 being the sample variances of the samples $(X_1, \dots, X_{n_1})$ and $(Y_1, \dots, Y_{n_2})$, respectively.

Proof. It follows from Theorem A.6.

Corollary 3.4. It follows from Propositions 3.2 and 3.3 that a level $1 - \alpha$ CI for σ_1^2/σ_2^2 is

$$\left(\frac{S_1^2}{S_2^2} \cdot \frac{1}{f_{\alpha/2}(n_1 - 1, n_2 - 1)}, \frac{S_1^2}{S_2^2} \cdot f_{\alpha/2}(n_2 - 1, n_1 - 1) \right).$$

If, say, $x_1, \dots, x_{20}$ and $y_1, \dots, y_{30}$ are realizations of the independent samples $X_1, \dots, X_{20} \sim N(\mu_1, \sigma_1^2)$ and $Y_1, \dots, Y_{30} \sim N(\mu_2, \sigma_2^2)$ then if $s_1^2 = 7, s_2^2 = 9$ then a 90% CI for σ_1^2/σ_2^2 is $(\frac{7^2}{9^2} \cdot \frac{1}{1.96}, \frac{7^2}{9^2} \cdot 2.07) = (0.30, 1.25)$.

Appendix

A

Theorem A.1. *If U is an open set in $\mathbb{R}^n$ and $g : U \rightarrow \mathbb{R}^n$ is a one-to-one and differentiable transformation, $B \subset U$, $f : g(B) \rightarrow \mathbb{R}$ then under some regularity conditions we have*

$$\int_{g(B)} f(y) dy = \int_B f(g(x)) |\det J_g(x)| dx$$

where $J_g(x)$ is the Jacobian of the transformation g at x , that is, $J_g(x) = \frac{\partial(g_1, \dots, g_n)}{\partial(x_1, \dots, x_n)}(x)$ in the sense that if either integral exists then so does the other one.

Theorem A.2. *If X is a random vector in $\mathbb{R}^n$ with probability density function $f_X(x)$ (where $x = (x_1, \dots, x_n)$) and g is one-to-one, moreover, g^{-1} is differentiable then $Y = g(X)$ has probability density as*

$$f_Y(x) = f_X(g^{-1}(x)) |\det J_{g^{-1}}(x)|, \quad (8)$$

Proof. If $B \subset \mathbb{R}^n$ then

$$\begin{aligned} \mathbb{P}(g(X) \in B) &= \mathbb{P}(X \in g^{-1}(B)) = \int_{g^{-1}(B)} f_X(x) dx = \\ &= \int_B f_X(g^{-1}(x)) |\det J_{g^{-1}}(x)| dx \end{aligned}$$

by Theorem A.1.

Theorem A.3. *If $X \sim N(0, \sigma^2 I_n)$ and $U \in \mathbb{R}^{n \times n}$ is orthogonal then the probability density function of $Y = UX$ is the same as that of X , therefore the marginals of Y are again independent $N(0, \sigma^2)$.*

Proof. By (8) the probability density $f_Y(y)$ of Y is (not taking the constants)

$$\exp\left(-\frac{1}{2}(U^{-1}y)^T(U^{-1}y)\right) \cdot |\det U^{-1}| = \exp\left(-\frac{1}{2}y^T y\right).$$

Definition A.4. *Let $Z_1, \dots, Z_n \sim N(0, 1)$ independent. Then the distribution of $\sum_{i=1}^n Z_i^2$ is called χ^2 (chi-square, or chi-squared) - distribution with n degrees of freedom and is denoted by $\chi^2(n)$.*

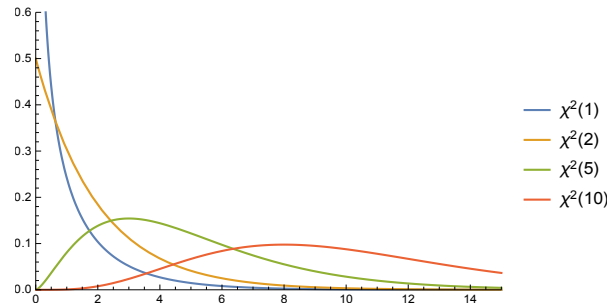


Figure 3: The densities of the χ^2 - distribution.

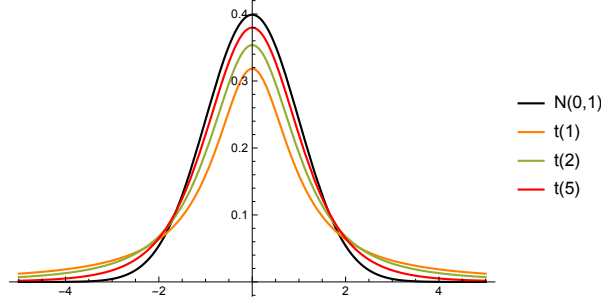


Figure 4: The densities of the t - distribution.

Definition A.5. Let $Z_0, Z_1, \dots, Z_n \sim N(0, 1)$ independent. Then the distribution of $\frac{Z_0}{\sqrt{\frac{1}{n} \sum_{i=1}^n Z_i^2}}$ is called $\mathbf{t}$ - distribution with n degrees of freedom and is denoted by $\mathbf{t}(n)$.

Theorem A.6. Let the sample $(X_1, \dots, X_n)$ have $N(\mu, \sigma^2)$ as a parent distribution. Then $\bar{X}_n$ and S^2 are independent and $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ (which is also denoted by $(n-1)S^2 \sim \sigma^2 \chi^2(n-1)$.)

Proof. We may suppose w.l.o.g. that $\mu = 0$. Let U be an $n \times n$ orthogonal matrix with the first row as $(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \dots, \frac{1}{\sqrt{n}})$. Then $Y = UX$ is $N(0, \sigma^2 I_n)$ and $Y_1 = \sqrt{n} \cdot \bar{X}_n$. Since $\sum_{i=1}^n Y_i^2 = \sum_{i=1}^n X_i^2$ therefore (using Steiner's identity) $n\bar{X}_n^2 + \sum_{i=2}^n Y_i^2 = (n-1)S^2 + n\bar{X}_n^2$, we have the statements of the theorem.

Theorem A.7. Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, independent. Then

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} \sim \mathbf{t}(n-1) \quad (9)$$

Proof. Since

$$\frac{\bar{X}_n - \mu}{S/\sqrt{n}} = \frac{\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2/\sigma^2}{n-1}}},$$

the theorem follows from Theorem A.6.

Theorem A.8. Let $(X_1, \dots, X_{n_1})$, $(Y_1, \dots, Y_{n_2})$ be independent samples with parent distributions $N(\mu_1, \sigma^2)$, $N(\mu_2, \sigma^2)$, respectively. Then

$$\frac{(\bar{X}_{n_1} - \bar{X}_{n_2}) - (\mu_1 - \mu_2)}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim \mathbf{t}(n_1 + n_2 - 2). \quad (10)$$

Proof. Since $(n_1 + n_2 - 2)S_p^2$ follows a $\sigma^2 \chi^2(n_1 + n_2 - 2)$ distribution, the theorem follows similarly as in the proof of Theorem A.7.